

MultiQC Report Summary – DSMZ Cell Line RNA-seq Analysis

Overview

Report Generated: August 5, 2025

Total Samples: 151

Tool Used: MultiQC v1.29

Metric Focus: Picard InsertSizeMetrics

Data Source: /mnt/beegfs/mnt/work/bioinfo/home/chu25/dsmz/results/picard

Project Context: RNA-seq quality control for DSMZ cell line classification

Key Findings

Insert Size Statistics

- Median Insert Sizes: Range from 250 bp to 606 bp
- Mean Insert Sizes: Range from 405 bp to 1,510 bp

Well-Performing Samples

Most samples (~80%) showed healthy insert size profiles:

- Typical Median Insert Size: ~250–350 bp

Examples: NALM_6, JEKO1, REH, Most of the NG-13640 series

Samples with Abnormally Large Insert Sizes

Some samples showed unusually large mean insert sizes ($\geq 1,000$ bp):

- NG-A0611_HL60_siCTR_c_libLAB6071: 1,510 bp
- NG-A0611_HL60_siCTR_b_libLAB6069: 1,426 bp
- NG-A0611_HL60_siTGIF1_c_libLAB6072: 1,358 bp
- NG-A0611_HL60_siTGIF1_a_libLAB6068: 1,181 bp
- TF_1 and NALM_1: $>1,100$ bp

Interpretation in Plain English

Think of insert sizes like puzzle pieces. If they're too big, they might not fit well into the final picture. Most of your samples are puzzle-perfect! A few just need a second look. ■

Recommendations

Immediate Actions

- Investigate high insert size samples
- Review library prep and fragmentation protocols
- Consider filtering outliers

Downstream Analysis Tips

- Use insert size plots
- Correlate with alignment/duplication metrics
- Evaluate impact on classification

Technical Notes

- MultiQC v1.29 with Picard InsertSizeMetrics
- 151 DSMZ cell line samples
- Source: /results/picard/
- Metric: Paired-end insert size from RNA-seq BAMs

Summary Table

Total Samples: 151

Samples with Optimal Insert Size: ~120 (80%)

Samples with Abnormal Insert Size: ~20 (13–15%)

Critical Outliers (>1000 bp mean): ~6 samples

Conclusion

The overall RNA-seq data quality is very good. Most samples show expected insert size distributions.

However, some samples show large insert sizes and should be flagged or reviewed before classification.

Next Steps

1. Visual inspection of insert size plots
2. Filtering strategy for flagged samples
3. Proceed with classification analysis
4. Document QC actions